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By Amit Katiyar (MCA-JNU)

1. A bag contains 5 red and 3 white marbles and another bag contains 2 red and 6 white marbles. 3 marbles are drawn at random from each bag. What is the probability that the marbles drawn are of same colours:

NIMCET 2007

(a)$\frac{55}{784}$ (b)$\frac{45}{784}$ (c)$\frac{55}{56}$ (d)None

2. A and B are independent events. The probability that both will occur is 1/8 and that neither of them occur is 3/8. If $P(A) > P(B)$. Find $P(B)$

NIMCET 2007

(a)1/2 (b)1/4 (c)3/4 (d)None

3. A, B and C are shooting with probabilities $\frac{3}{5}, \frac{2}{5}, \frac{3}{4}$ respectively. The probability that the target is hit by at least two shots is:

NIMCET 2007

(a)0.63 (b)0.69 (c)0.18 (d)0.82

4. A random variable X has the probability distribution:

X:	1	2	3	4	5	6	7	8
$P(X)$	0.15	0.23	0.12	0.10	0.20	0.08	0.07	0.05

For the events E = {X is a prime number} and $F = \{X < 4\}$, the proability $P(E \cup F)$ is: NIMCET 2007

(a)0.35 (b)0.77 (c)0.87 (d)0.50

5. A bag contains 5 red and 3 white marbles and another bag contains 2 red and 6 white marbles. 2 marbles are drawn at random from each bag. What is the probability that the marbles drawn are of same colours: (Probability)

NIMCET 2007

(a)$\frac{55}{784}$ (b)$\frac{45}{784}$ (c)$\frac{55}{56}$ (d)none

6. A and B are independent events. The probability that both will occur is 1/8 and that neither of them occur is NIMCET 2007
3/8. If P(A) > P(B) . Find P(B) (probability)
(a)1/2 (b)1/4 (c)3/4 (d)none

7. A, B and C are shooting with probabilities $\frac{3}{5}, \frac{2}{5}, \frac{3}{4}$ respectively. The probability that the target is hit by at least two shots is : (Probability) NIMCET 2007

(a)0.63 (b)0.69 (c)0.18 (d)0.82

8. A random variable X has the probability distribution :

X : 1 2 3 4 5 6 7 8
P(X) : 0.15 0.23 0.12 0.10 0.20 0.08 0.07 0.05
For the events E={ X is a prime number } and
F={ X<4}, the probability P(E ∪ F) is : (Probability)

NIMCET 2007

(a)0.35 (b)0.77 (c)0.87 (d)0.50

9. If two events A and B are such that $P(A') = 03, P(B) = 05$ and $P(A \cap B) = 03$, then $P\left(\frac{B}{A \cup B'}\right)$is NIMCET 2008

(a)$\frac{1}{4}$ (b)$\frac{3}{8}$ (c)$\frac{1}{8}$ (d)None

10. A pair of unbiased dice is rolled together till a sum of either 5 or 7 is obtained. The probability that 5 comes befor 7 is

NIMCET 2008

(a)$\frac{3}{5}$ (b)$\frac{2}{5}$ (c)$\frac{4}{5}$ (d)None

11. A letter is taken at random from the letters of the word 'STATISTICS' and another letter is taken at random from the letters of the word "ASSISTANT'. The probability that all the balls selected are red is NIMCET 2008

(a)$\frac{1}{45}$ (b)$\frac{13}{90}$ (c)$\frac{19}{90}$ (d)$\frac{5}{8}$

12. A bag contains 6 red and 4 green balls. A fair dice is rolled and number of balls equal to the appearing on the dice is chosen from the urn at random. The probability that all the balls selected are red is NIMCET 2008

(a)$\frac{1}{3}$ (b)$\frac{3}{10}$ (c)$\frac{1}{8}$ (d)None

13. A six faced dice is biased one. It is thrice more likely to show an odd number that to show an even number. If is thrown twice. The probability that the sum of the numbers in the two throws is even, is NIMCET 2008

(a)$\frac{4}{8}$ (b)$\frac{5}{8}$ (c)$\frac{6}{8}$ (d)$\frac{7}{8}$

14. A letter is known to have come from either TATANAGR or CALCUTTA. On the envelope, just two consecutive letters. TA, are visible. The probability that the letter has come from CALCUTTA is

NIMCET 2008

(a)$\frac{4}{11}$ (b)$\frac{1}{3}$ (c)$\frac{5}{12}$ (d)None

15. A man has 5 coins, two of which are double-headed, one is double-tailed and two are normal. He shuts his eyes, picks a coin at random and tosses it. The probability that the lower face of the coin is head, is

NIMCET 2009

(a)$\frac{1}{5}$ (b)$\frac{2}{5}$ (c)$\frac{3}{5}$ (d)$\frac{4}{5}$

16. A and B are independent witness in a case. The probability that A speaks the truth is 'x' and that B speaks the truth is 'y'. if A and B agree on a certain statement, the probability that the statement is true, is

NIMCET 2009

(a)$\frac{xy}{xy+(1-x)(1-y)}$ (b)$\frac{xy}{(1-x)(1-y)}$
(c)$\frac{(1-x)(1-y)}{xy+(1-x)(1-y)}$ (d)$\frac{(1-x)(1-y)}{xy}$

17. Let A and B be two events such that $P(\overline{A \cup B}) = \frac{1}{6}$, $P(A \cap B) = \frac{1}{4}$ and $P(\overline{A}) = \frac{1}{4}$. Then events A and B are

NIMCET 2009

(a) independent but not equally likely
(b) mutually exclusive and independent
(c) equally likely and mutually exclusive
(d) equally likely but not independent

18. A and B thrown a dice in succession to win a bet with A starting first. Whoever throws '1' first wins an amount of Rs. 110. What are the respective expectations to A and B?

NIMCET 2009

(a) Rs 70 and Rs 40 (b)Rs 60 and Rs 50
(c)Rs 75 and Rs 35 (d) None of these

19. The probability that a man who is 85 yr. old will die before attaining the age of 90 yr. is 1/3, A_1, A_2, A_3 and A_4 are four persons who are 85 yr. old, The probability that A_1 will die before attaining the age of 90 yr and will be the first to die is **NIMCET 2009**
(a) 65/81 (b)13/81 (c)65/324 (d)13/108

20. An anti-aircraft gun can take a maximum of four shots at an enemy plane moving away from it. The probabilities of hitting the plane at first, second, third and fourth shots are 0.4, 0.3, 0.2 and 0.1 respectively. The probability that the gun hits the plane then is **NIMCET 2009**
(a) 0.6972 (b)0.6978 (c)0.6976 (d)0.6974

21. India plays two matches each with West Indies and Ausstralia. In any match the probabilities of India getting point 0,1 and 2 are 0.45, 0.05 and 0.50 respectively. Assuming that the outcomes are independent, the probability of India getting at least 7 points is **NIMCET 2010**
(a) 0.8750 (b)0.0875 (c)0.0625 (d)0.0250

22. A coin is tossed three times. The probabilities of getting head and tail alternatively is **NIMCET 2010**
(a)1/11 (b)2/3 (c)3/4 (d)1/4

23. One hundred indentical coins, each with probability p of showing up a head, are tossed. If $0 < p < 1$ and if the probability of heads on exactly 50 coins is equal to that of heads on exactly 51 coins then the value of p, is **NIMCET 2010**
(a)1/2 (b)49/101 (c)50/101 (d)51/101

24. In a Possion distribution if $P[X = 3] = \frac{1}{4}P[X = 4]$ then $P[X = 5] = kp[X = 7]$ where 'k' equal to **NIMCET 2010**
(a)1/7 (b)21/4 (c)21/128 (d)21/256

25. A random variable X has the following probability distribution
Then, the value of 'a' is

X	0	1	2	3	4	5	6	7	8
P(X=x)	a	3a	5a	7a	9a	11a	13a	15a	17a

NIMCET 2011
(a)1/81 (b)2/82 (c)5/82 (d)7/81

26. An anti-aircraft gun can take a maximum of four shots at an enemy plane moving away from it. The probabilities of hitting the plane at the first, second, third and fourth shots are 0.4, 0.3, 0.1 and 0.1 respectively. The probability that the gun hit the plane, then is **NIMCET 2011**
(a)0.5 (b)0.7235 (c) 0.6976 (d)0.1

27. The probability of a razor blade to be defective is 0.002. the blades are in packet of 10. The number of packets containing no defective blade in a stock of 10000 packets is **NIMCET 2011**
(a)2000 (b)9980 (c)9950 (d)8000

28. A determinant is chosen at random from the set of all determinants of matrices of order 2 with elements 0 and 1 only. The probability that the determinant chosen is non-zero is **NIMCET 2012**
(a)$\frac{3}{16}$ (b)$\frac{3}{8}$ (c)$\frac{1}{4}$ (d)None

29. Coefficients of quadratic equation $ax^2 + bx + c = 0$ are choosen by tossing there fair coins, where 'head' means one and 'tall' means two. Then the probability that roots of the equtionare imaginary, is **NIMCET 2012**
(a)$\frac{7}{8}$ (b)$\frac{5}{8}$ (c)$\frac{3}{8}$ (d)$\frac{1}{8}$

30. Let P(E) denote the probability of event E. Given $P(A) = 1, P(B) = \frac{1}{2}$, the value of $P(A|B)$ and $P(B|A)$ respectively are **NIMCET 2012**
(a)$\frac{1}{4}, \frac{1}{2}$ (b)$\frac{1}{2}, \frac{1}{4}$ (c)$\frac{1}{2}, 1$ (d)$1, \frac{1}{2}$

31. If a fair coin is tossed n times, then the probability that the head comes odd number of times is **NIMCET 2012**
(a)$\frac{1}{2}$ (b)$\frac{1}{2^n}$ (c)$\frac{1}{2^{n-1}}$ (d)None

32. One hundred identical coins each with probability P of showing up heads are tossed. If $0 < P < 1$ and the probability of heads showing on 50 coins is equal to that of heads 51 coins, then the value of P is **NIMCET 2012**
(a)$\frac{1}{2}$ (b)$\frac{49}{101}$ (c)$\frac{50}{101}$ (d)$\frac{51}{101}$

33. A problem in Mathematics is given to three students A, B and C whose chances of solving it are $\frac{1}{2}, \frac{1}{3}$ and $\frac{1}{4}$ respectively. If they all try to solve the problem, what is the probability that the problem will be solved? **NIMCET 2012**
(a)$\frac{1}{2}$ (b)$\frac{1}{4}$ (c)$\frac{1}{3}$ (d)$\frac{3}{4}$

34. A six faced dice is a biased one. It is thrice more likely to show an odd number than to show an even number. It is thrown twice, the probability that the sum of the numbers in the two throws is even, is **NIMCET 2008, 2013**
(a)$\frac{4}{8}$ (b)$\frac{5}{8}$ (c)$\frac{6}{8}$ (d)$\frac{7}{8}$

35. Person A can hit a target 4 times in 5 attempts. Person B can hit a target 3 times in 4 attempts. Person C can hit a target 2 times in 3 attempts. They fire a volley. The probability that the target is hit atleast two times, is **NIMCET 2013**
(a)$\frac{3}{4}$ (b)$\frac{1}{2}$ (c)$\frac{5}{6}$ (d)1

36. Atal speaks truth in 70% and George speaks the truth in 60% cases. In what percentage of cases they are likely to contradict each other in stating the same fact? **NIMCET 2013**
(a)$\frac{13}{50}$ (b)$\frac{11}{50}$ (c)$\frac{27}{50}$ (d)$\frac{23}{50}$

37. If A and B are two events such that $P(A \cup B) = \frac{5}{6}$, $P(A \cap B) = \frac{1}{3}$ and $P(\overline{B}) = \frac{1}{2}$ then the events A and B are

NIMCET 2013

(a) dependent (b) independent
(c)mutually exclusive (d)None of these

38. Two fair dice are tossed. What is the probability that the total score is a prime number?

NIMCET 2013

(a)$\frac{1}{6}$ (b)$\frac{5}{12}$ (c)$\frac{1}{2}$ (d)$\frac{7}{9}$

39. An experiment succeeds twice often as it fails. The probability that in the next six trails there will be atleast four successes, is

NIMCET 2013

(a)$\frac{240}{729}$ (b)$\frac{496}{729}$ (c)$\frac{220}{729}$ (d)$\frac{223}{729}$

40. A chain of video stores sale three different brands of DVD players. Of its DVD player sales 50% are brand 1, 30% are brand 2 and 20% are brand 3. Each manufacture offers one year warranty on parts and labour. It is known that 25% of brand 1 DVD players require warranty repair work, whereas the corresponding percentage for brand 2 and 3 are 20% and 10% respectively. The probability that a randomly selected purchaser has a DVD player that will need repair while under warranty, is **NIMCET 2014**

(a)0.795 (b)0.205 (c)0.1250 (d)0.060

41. Suppose r integers, $0 < r < 10$, are chosen from (0, 1, 2, ….., 9) at random and with replacement. The probability that no two are equal, is

NIMCET 2014

(a)$\frac{10!}{10!r!}$ (b)$\frac{10!}{10!(10-r)}$
(c)$\frac{10!}{r!(10-r)!}$ (d)$\frac{10}{10^r \times r! \times (10-r)!}$

42. A box contains 3 coins, one coin is fair, one coin is two headed and one coin is weighted, so that the probability of heads appearing is $\frac{1}{3}$. A coin is selected at random and tossed, then the probability that head appears, is

NIMCET 2014

(a)$\frac{11}{18}$ (b)$\frac{7}{18}$ (c)$\frac{1}{8}$ (d)$\frac{1}{4}$

43. A student takes quiz consisting of 5 multiple choice questions. Each question has 4 possible answer. If a student is guessing the answer at random and answer to different are independent, then the probability of atleast one correct answer is

NIMCET 2014

(a)0.237 (b)0.00076 (c)0.7623 (d)1

44. A is targeting B, B and C are targeting A. probabilities of hitting the target by A, B and C are $\frac{2}{3}, \frac{1}{2}$ and $\frac{1}{2}$ respectively. If A is hit, then the probability that B hits the target and C does not, is

NIMCET 2015

(a)$\frac{1}{2}$ (b)$\frac{1}{3}$ (c)$\frac{2}{3}$ (d)$\frac{3}{4}$

45. Suppose that, A and B are two events with probabilities $P(A) = \frac{1}{2}, P(B) = \frac{1}{3}$. Then, which of the following is true?

NIMCET 2015

(a)$\frac{1}{3} \le P(A \cap B) \le \frac{1}{2}$ (b)$\frac{1}{4} \le P(A \cap B) \le \frac{1}{3}$
(c)$\frac{1}{6} \le P(A \cap B) \le \frac{1}{3}$ (d)$\frac{1}{4} \le P(A \cap B) \le \frac{1}{2}$

46. If a fair dice is rolled successively, then the probability that 1 appears in an even numbered throw is

NIMCET 2015

(a)$\frac{5}{36}$ (b)$\frac{3}{11}$ (c)$\frac{1}{6}$ (d)$\frac{5}{11}$

47. Out of $2n+1$ tickets, which are consecutively numbered, three are drawn at random. Then, the probability that, the numbers on them are in arithmetic progression is

NIMCET 2015

(a)$\frac{n^2}{4n^2-1}$ (b)$\frac{n}{4n^2-1}$ (c)$\frac{3n^2}{4n^2-1}$ (d)$\frac{3n}{4n^2-1}$

48. A box contains 2 blue caps, 4 red caps, 5 green caps and 1 yellow caps. If four caps are picked at random, then the probability that none of them is green, is

NIMCET 2016

(a)$\frac{7}{99}$ (b)$\frac{7}{12}$ (c)$\frac{5}{99}$ (d)$\frac{5}{12}$

49. An experiment has 10 equally likely outcomes. Let A and B be two non-empty events of the experiment. If A consists of 4 outcomes, the number of outcomes that B must have so that A and B are independent, is

NIMCET 2016

(a)2, 4 or 8 (b)3, 6 or 9 (c)4 or 8 (d)5 or 10

50. For any two events A and B , the probability that atleast one of them occur is 0.6. if A and B occurs simultaneously with a probability 0.3, then $P(A') + P(B')$ is

NIMCET 2016

(a)0.9 (b)1.15 (c)1.1 (d)1.0

51. The probability that A speaks truth is 4/5 white the probability for B is 3/4 . the probability that they contradict each other when asked to speak on a fact is **NIMCET 2016**

(a)3/20 (b)1/5 (c)7/20 (d)4/5

52. Three houses are available in a locality. Three perons apply for the houses. Each applies for one house without consulting other. The probability that all the three apply for the same houses is **NIMCET 2016**

(a)$\frac{8}{9}$ (b)$\frac{1}{5}$ (c)$\frac{2}{5}$ (d) $\frac{1}{9}$

53. Five horses are in a race. Mr. A selects two of the horses at random and bets on them. The probability that Mr. A selected the winning horses is **NIMCET 2016**

(a)$\frac{3}{5}$ (b)$\frac{1}{5}$ (c)$\frac{2}{5}$ (d)$\frac{4}{5}$

54. A and B are independent witness in a case. The chance that A speaks truth is x and B speaks truth is y. if A and B agree on certain statement, the probability that the statement is true, is

NIMCET 2017

(a)$\frac{xy}{xy+(1-x)(1-y)}$ (b)$\frac{xy}{(1-x)(1-y)}$
(c)$\frac{(1-x)(1-y)}{xy+(1-y)(1-x)}$ (d)$\frac{x+y}{xy(1-x)(1-y)}$

55. In an entrance test, there are multiple choice questions, with four possible answers to each question of which one is correct. The probability that a student knows the answer to a question is 90%. If the student gets the correct answer to a question, then the probability that he was guessing is **NIMCET 2017**

(a)$\frac{37}{40}$ (b)$\frac{1}{37}$ (c)$\frac{36}{37}$ (d)$\frac{1}{9}$

56. A man is known to speak the truth 2 out of 3 times. He threw a dice cube with 1 to 6 on its faces and report that it is 1. Then the probability that it is actually 1 is **NIMCET 2017**

(a)$\frac{1}{2}$ (b)$\frac{1}{7}$ (c)$\frac{2}{7}$ (d)$\frac{5}{6}$

57. Let A and B be two events such that $P(\overline{A\cup B}) = \frac{1}{6}$, $P(A\cap B) = \frac{1}{4}$ and $P(\overline{A}) = \frac{1}{4}$, where $\overline{A}$ stands for complement of event A. then the events A and B are **NIMCET 2017**

(a) independent but not equally likely
(b) mutually exclusively and independent
(c) equally likely and mutually
(d) equally likely but not independent

58. If E_1 and E_2 are two events associated with a random experiment such that $P(E_2) = 0.35$, $P(E_1 \; or \; E_2) = 0.85$ and $P(E_1 \; and \; E_2) = 0.15$, then $P(E_1)$ is **NIMCET 2017**

(a)0.25 (b)0.35 (c)0.65 (d)0.75

59. The mean and variance of a random variable X having binomial distribution are 4 and 2, respectively. Then $P(X = 1)$ is **NIMCET 2017**

(a)$\frac{1}{32}$ (b)$\frac{1}{16}$ (c)$\frac{1}{8}$ (d)$\frac{1}{4}$

60. If A and B are two events and $P(A\cup B) = 5/6$, $P(A\cap B) = 1/3$, $P(B) = \frac{1}{2}$, then B are two events which are **NIMCET 2018**

(a) dependent (b) independent
(c) mutually exclusive (d) equally likely

61. Two persons A and B agree to meet on 20th April, 2018 between 6 P.M. and 7 P.M. with the understanding that they will wait no longer than 20 minutes for the other. What is the probability that they meet? **NIMCET 2018**

(a)5/9 (b)7/9 (c)2/9 (d)4/9

62. Three numbers a, b and c are chosen at random (without replacement) from among the numbers 1, 2, 3......99. The probability that $a^3+b^3+c^3-3abc$ is divisible by 3, is **NIMCET 2018**

(a)$\frac{3\cdot{}^{33}C_3+({}^{33}C_1)^3}{{}^{99}C_3}$ (b)$\frac{3\cdot{}^{33}C_3-({}^{33}C_1)^3}{{}^{99}C_3}$
(c)$\frac{2\cdot{}^{33}C_3+({}^{33}C_1)^3}{{}^{99}C_3}$ (d)$\frac{2\cdot{}^{33}C_3-({}^{33}C_1)^3}{{}^{99}C_3}$

63. A and B play a game where each is asked to selected a number form 1 to 25. If two numbers match, both of them win a prize. The probability that they will not win a prize. The probability that they will not win a prize in a single trial, is **NIMCET 2018**

(a)$\frac{1}{25}$ (b)$\frac{24}{25}$ (c)$\frac{2}{25}$ (d)None

64. Two numbers a and b are chosen at random form a set of first 30 natural numbers, then the probability that a^2-b^2 is divisible by 3 is **NIMCET 2019**

(a)$\frac{47}{87}$ (b)$\frac{15}{87}$ (c)$\frac{12}{87}$ (d)$\frac{9}{87}$

65. The man a step forward with probability 0.4 and backwards with probability 0.6. the probability that at the end of eleven steps, he is one step away form the starting point is **NIMCET 2019**

(a)$462(0.32)^5$ (b)$462(0.04)^5$
(c)$462(0.14)^5$ (d)$462(0.24)^5$

66. Let U and V be two events of a sample space S and P(A) denote the probability of an event A. which of the following statements is true? **NIMCET 2019**

(a) If $P(U) = P(V)$ then $U = V$
(b) If $P(U) = 0$ then $U^c = S$
(c) If $U\cap V = \phi$ then U and V are independent
(d) If U and V are independent, then U^c and V^c are also independent.

67. If a man purchases a raffic ticket, he can win a first prize of Rs. 5000 or a second prize of Rs. 2000 with probabilities 0.001 and 0.003 respectively. What should be a fair price to pay for the ticket? **NIMCET 2019**

(a) Rs 11 (b) Rs 15 (c)Rs 2000 (d)None

68. A computer producing factory has only two plants T_1 and T_2. Plant T_1 produces 20% and plant T_2 produces 80% of the total computers produced 7% of the computers produced in the factory turn out to be defective. It is known that P(computer turns out to be defective given that it is produced in plant T_1) = 10 P (computer turns out to be defective given that it is produces in plant T_2). A computer produced in the factory is randomly selected and it does not turn out to be defective. Then the probability that it is produced in plant T_2 is **NIMCET 2019**

(a)$\frac{36}{73}$ (b)$\frac{47}{79}$ (c)$\frac{78}{93}$ (d)$\frac{75}{83}$

69. A problem im Mathematics is given to 3 students A, B and C. if the probability of A solving the problem is $\frac{1}{2}$ and B not solving it is $\frac{1}{4}$. The whole probability of the problem being solved is $\frac{63}{64}$, then what is the probability of solving it by C? **NIMCET 2020**

(a)$\frac{1}{8}$ (b)$\frac{1}{64}$ (c)$\frac{7}{8}$ (d)$\frac{1}{2}$

70. A and B play a game where each is asked to select a number form 1 to 25. If the two numbers match, both win a prize. The probability that they will not win a prize in a single trial is **NIMCET 2020**

(a)$\frac{1}{25}$ (b)$\frac{24}{25}$ (c)$\frac{2}{25}$ (d)$\frac{3}{25}$

71. If three throws of three dices, the probability of throwing triplets not more than twice is

NIMCET 2021

(a) $1-\frac{1}{6^2}$ (b) $1-\frac{1}{6^3}$ (c) $1-\frac{1}{36^2}$ (d) $1-\frac{1}{36^3}$

72. The probability of occurrence of two events E and F are 0.25 and 0.50, respectively. The probability of their simultaneous occurrence is 0.14. the probability that neither E nor F occurs is

NIMCET 2021

(a)0.61 (b)0.11 (c)0.39 (d)0.89

73. The probability that a man who is x years old will die in a year is p. then, amongst n persons $A_1, A_2 \ldots . A_n$ each x years old now, the probability that A_1 will die in one year os

NIMCET 2021

(a) $\frac{1}{n^2}$ (b) $1-(1-P)^n$
(c) $\frac{1}{n^2}[1-(1-P)^n]$ (d) $\frac{1}{n}[1-(1-P)^n]$

74. If a number x is selected from natural numbers 1, 2... 100, then the probability for $x+\frac{100}{x}>29$ is

NIMCET 2021

(a) $\frac{39}{50}$ (b) $\frac{43}{50}$ (c) $\frac{41}{50}$ (d) $\frac{37}{50}$

75. There are two circles in xy- plane whose equations are $x^2+y^2-2y=0$ and $x^2+y^2-2y-2=0$. A point (x, y) is chosen at random inside the larger circle. Then the probability that the points has been taken form smaller circle is

NIMCET 2022

(a)1/3 (b)2/3 (c)1/2 (d)1/4

76. If $0<P(A)<1$ and $0<P(B)<1$ and $P(A\cap B)=P(A)P(B)$, then

NIMCET 2022

(a) $P(B|A)=P(B)-P(A)$
(b) $P(A^c-B^c)=P(A^c)-P(B^c)$
(c) $P(A\cup B)^c=P(A^c)P(B^c)$
(d) $P(A|B)=P(A)-P(B)$

77. A four digit number is formed using the digits 1, 2, 3, 4, 5 without repetition. The probability that it is divisible by 3 is

NIMCET 2022

(a)1/3 (b)1/4 (c)1/5 (d)1/6

78. A computer producing factory has only two plants T_1 and T_2. Plant T_1 produces 20% and plant T_2 produces 80% of the total computers produced 7% of the computers produced in the factory turn out to be defective. It is known that P(computer turns out to be defective given that it is produced in plant T_1) = 10 P (computer turns out to be defective given that it is produces in plant T_2). A computer produced in the factory is randomly selected and it does not turn out to be defective. Then the probability that it is produced in plant T_2 is

NIMCET 2023

(a) $\frac{36}{73}$ (b) $\frac{47}{79}$ (c) $\frac{78}{93}$ (d) $\frac{75}{83}$

79. A speaks truth is 60% and B speaks the truth in 50% cases. In what percentage of cases they are likely in contradict each other while narrating some incident is

NIMCET 2023

(a)1/2 (b)1/4 (c)2/3 (d)1/3

80. Given two events A and B such that idd in favour A are 2 : 1 and odd in favour of $A\cup B$ are 3 : 1. Consistent with this information the smallest and largest value for the probability of vent B are given by

NIMCET 2023

(a) $\frac{1}{12}\le P(B)\le\frac{3}{4}$ (b) $\frac{1}{3}\le P(B)\le\frac{1}{2}$
(c) $\frac{1}{6}\le P(B)\le\frac{1}{3}$ (d)None of these

81. A bag contain different kind of balls in which 5 yellow, 4 black and 3 green balls. If 3 balls are drawn at random then find the probability that no black ball is chosen.

NIMCET 2023

(a) $\frac{14}{55}$ (b) $\frac{1}{66}$ (c) $\frac{2}{9}$ (d)None

82. Box A contains 4 green, 5 black and 6 white balls. Box B contains 5 green, 3 black and 2 white balls.
A ball is chosen at random from the box A and placed in Box B and the a ball is chosen at random from box B. what is the probability that a black ball was transferred from box A to box B given that the final ball chosen from box B is green?

NIMCET 2023

(a)30/79 (b)25/79 (c)24/79 (d)None

83. Region R is defined as region in first quadrant satisfying the condition $x^2 + y^2 < 4$. Given that a point P = (r, s) lies in R, what is the probability that r > s? (Probability)

(a) 1 (b) 0
(c) ½ (d) 1/3

NIMCET 2024

84. A coin is thrown 8 number of times. What is the probability of getting a head in an odd number of throw? (Probability)

(a) ¾ (b) ¼
(c) ½ (d) 1/8

NIMCET 2024

85. A critical orthopedic surgery is performed on 3 patients. The probability of recovering a patient is 0.6. Then the probability that after surgery, exactly two of them will recover is (Probability)

(a) 0.123 (b) 0.432
(c) 0.321 (d) 0.234

NIMCET 2024

86. If three distinct numbers are chosen randomly from the first 100 natural numbers, then the probability that all three of them are divisible by both 2 and 3 is (Probability)

(a) 4/35 (b) 4/33
(c) 4/1155 (d) 4/25

NIMCET 2024

87. Let A and B be two events defined on a sample space Ω. Suppose A^C the complement of A relative to the sample space Ω. Then the probability $P((A \cap B^C) \cup (A^C \cap B))$ equal (probability) **NIMCET 2024**
(a) $P(A) + P(B) + P(A \cap B)$
(b) $P(A) + P(B) - P(A \cap B)$
(c) $P(A) + P(B) + 2P(A \cap B)$
(d) $P(A) + P(B) - 2P(A \cap B)$

88. A speaks truth in 40% and B in 50% of the cases. The probability that they contradict each other while narrating some incident is **NIMCET 2024**
(a) 2/3 (b) ½
(c) 1/3 (d) 0

89. The captains of five cricket teams, including India and Australia, are lined up randomly next to one other for a group photo. What is the probability that the captains of India and Australia will stand next to each other? **NIMCET 2025**
(a) $\frac{1}{4}$ (b) $\frac{2}{5}$
(c) $\frac{1}{2}$ (d) $\frac{1}{5}$

90. Let E and F be two events such that P(E) > 0 and P(F) > 0. Which one of the following is **NOT** equivalent to the condition that P(E) = P(E/F)? **NIMCET 2025**
(a) P(F) = P(F/E)
(b) E and F are independent
(c) E^c and F are independent
(d) $P(E^c)\,P(F^c) \neq P(E^c \cap F^c)$

91. Consider the sample space) $\Omega = \{(x, y): x, y \in \{1,2,3,4\}\}$ where each outcome is equally likely. Let $A = \{x \geq 2\}$ and $B = \{y > x\}$ be two events. Then which of the following is **NOT** true? **NIMCET 2025**
(a) $P(A \cap B) = 1/4$
(b) A and B are not independent
(c) P(B)=3/8
(d) P(A) = 3/4

92. The scores of students in a national level examination are normally distributed with a mean of 500 and a standard deviation of 100. If the value of the cumulative distribution of the standard normal random variable at 0.5 is 0.691, then the probability that a randomly selected student scored between 450 and 500 is **NIMCET 2025**
(a) 0.191 (b) 0.391
(c) 0.591 (d) 0.091

93. The number of accidents per week in a town follows Poisson distribution with mean 2. If the probability that there are three accidents in two weeks time is ke^{-6}, then the value of k is **NIMCET 2025**
(a) 18 (b) 36
(c) 27 (d) 9 .

94. There are two coins, say blue and red. For blue coin, probability of getting head in 0.99 and for red coin, it is 0.01. One coin is chosen randomly and is tossed. The probability of getting head is **NIMCET 2025**
(a) 1 (b) 0.98 (c) 0.5 (d) 0.02

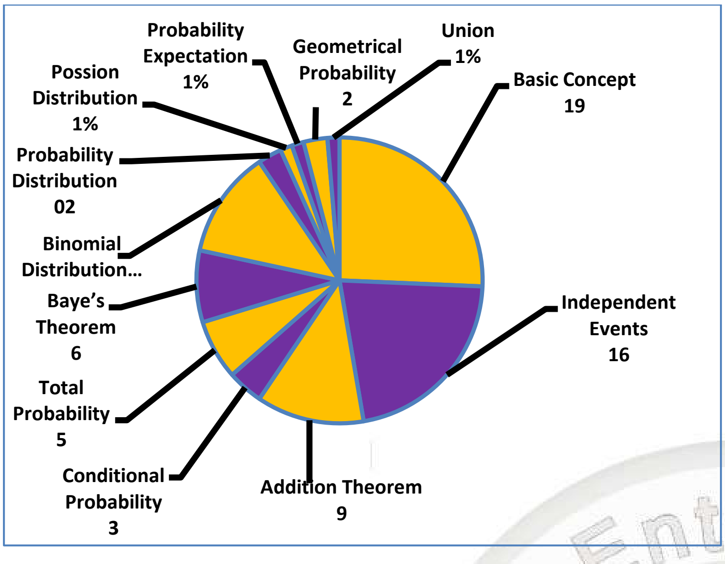
ANSWER KEY

1.	a	2.	b	3.	a	4.	b	5.	a
6.	b	7.	a	8.	b	9.	b	10.	b
11.	c	12.	d	13.	b	14.	a	15.	c
16.	a	17.	a	18.	b	19.	c	20.	c
21.	b	22.	d	23.	d	24.	c	25.	a
26.	c	27.	b	28.	b	29.	a	30.	d
31.	a	32.	d	33.	d	34.	b	35.	c
36.	*	37.	b	38.	b	39.	b	40.	b
41.	d	42.	a	43.	c	44.	a	45.	c
46.	d	47.	d	48.	a	49.	d	50.	c
51.	c	52.	d	53.	c	54.	a	55.	b
56.	c	57.	a	58.	c	59.	a	60.	b
61.	a	62.	a	63.	b	64.	a	65.	d
66.	b, d	67.	a	68.	c	69.	c	70.	b
71.	d	72.	c	73.	d	74.	a	75.	d
76.	c	77.	c	78.	c	79	a	80	a
81	a	82	b	83.	c	84.	c	85.	b
86.	c	87.	d	88.	b	89.	b	90.	d
91.	a	92.	a	93.	b	94.	c		

Questions Asked in NIMCET 2008 to 2023

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	08	NIMCET 2016	06
NIMCET 2008	06	NIMCET 2017	06
NIMCET 2009	06	NIMCET 2018	04
NIMCET 2010	04	NIMCET 2019	05
NIMCET 2011	03	NIMCET 2020	02
NIMCET 2012	06	NIMCET 2021	04
NIMCET 2013	06	NIMCET 2022	03
NIMCET 2014	04	NIMCET 2023	05
NIMCET 2015	04	NIMCET 2024	06
		NIMCET 2025	06

SUB TOPICWISE ANALYSIS



SUB TOPICWISE QUESTIONS ASKED IN NIMCET

Sub Topic	Ques. Asked	Sub Topic	Ques. Asked
Basic Concept	19	Binomial Distribution	09
Independent Events	16	Probability Distribution	02
Addition Theorem	09	Possion Distribution	01
Conditional Probability	03	Probability Expectation	01
Total Probability	05	Geometrical Probability	02
Baye's Theorem	06	Union	01

SOLUTION

1. (a)

(a) $\begin{vmatrix} 5 & R \\ 3 & W \end{vmatrix}$ $\begin{vmatrix} 2 & R \\ 6 & W \end{vmatrix}$

I II

Case-I: Red marbles drawn from each box

$P(E) = \frac{{}^5C_2}{{}^8C_2} \times \frac{{}^2C_2}{{}^8C_2} = \frac{10}{(28)^2}$

Case-II: If white marbles draw

$P(E) = \frac{{}^3C_2}{{}^8C_2} \times \frac{{}^6C_2}{{}^8C_2} = \frac{3\times15}{(28)^2}$

Total $= \frac{10}{784} + \frac{45}{784} = \frac{55}{784}$

2. (b)

The probability of getting an odd number of heads is $\frac{1}{2}$.

Step 1:

Express P $(A \cup B)$ in term of P(A) and P(B):

P $(\bar{A} \cap \bar{B}) = 1 - P(A \cup B)$

$\frac{3}{8} = 1 - P(A \cup B)$

$P(A \cup B) = 1 - \frac{3}{8} = \frac{5}{8}$

P $(A \cup B)$ = P (a)+P(B) –P $(A \cap B)$

$\frac{5}{8} = P(A) + P(B) - \frac{1}{8}$

P(A) + P (B) $= \frac{6}{8} = \frac{3}{4}$

Step 2

P $(A \cap B)$ =P (A) P (B)

$\frac{1}{8}$ = P (A) P (B)

Step 3

Let x = P (A) and y = P (B)

$x + y = \frac{3}{4}$

$xy = \frac{1}{8}$

$x = \frac{3}{4} - y$

$\left(\frac{3}{4} - y\right) y = \frac{1}{8}$

$\frac{3}{4}y - y^2 = \frac{1}{8}$

$y^2 - \frac{3}{4}y + \frac{1}{8} = 0$

$8y^2 - 6y + 1 = 0$

$y = \frac{6 \pm \sqrt{36-32}}{16}$

$y = \frac{6 \pm 2}{16}$

$y = \frac{1}{2}$, $x = \frac{1}{4}$, but this contradicts

P(A) > P(B).

If $y = \frac{1}{4}$, $x = \frac{1}{2}$. This satisfies P(A) > P(B).

The probability of even B is $\frac{1}{4}$.

3. (a)

$P(A) = \frac{3}{5}, P(B) = \frac{2}{5}, P(C) = \frac{3}{4}$

P(atleast hit by 2 shot) = P(A).P(B).P($\bar{C}$) +

P(A). P($\bar{B}$).P (C) +P($\bar{A}$).P (B).P (C) + P (A).P(B).P (C)

$= \frac{3}{5} \times \frac{2}{5} \times \frac{1}{4} + \frac{3}{5} \times \frac{3}{5} \times \frac{3}{4} + \frac{2}{5} \times \frac{2}{5} \times \frac{3}{4} + \frac{3}{5} \times \frac{2}{5} \times \frac{3}{4}$

$= \frac{6+27+12+18}{100}$

$= \frac{63}{100} = 0.63$

4. (b)

x =	1	2	3	4	5	6	7	8
P(x)	0.15	0.23	0.12	0.10	0.20	0.08	0.07	0.05

E { x is a prime no.)

F = {x < 4}, P (E U F) = ?

E = {2,3,5,7}

F = {x: 1,2,3}

E U F = {1,2,3,5,7}

P (E U F) = 1 – {P(x = 4)+ P (x = 6)+ P (x = 8)}

= 1 – (0.10 + 0.08 + 0.05)

1 – (0.23)

0.77

5. **(a)** $\begin{vmatrix} 5 & R \\ 3 & W \end{vmatrix}$ $\begin{vmatrix} 2 & R \\ 6 & W \end{vmatrix}$

I II

Case-I: Red marbles drawn from each box

$P(E) = \frac{{}^5C_2}{{}^8C_2} \times \frac{{}^2C_2}{{}^8C_2} = \frac{10}{(28)^2}$

Case-II: If white marbles draw

$P(E) = \frac{{}^3C_2}{{}^8C_2} \times \frac{{}^6C_2}{{}^8C_2} = \frac{3\times 15}{(28)^2}$

Total $= \frac{10}{784} + \frac{45}{784} = \frac{55}{784}$

6. **(b)** P(A ∩B) = $\frac{1}{8}$

P(A'∩B') = $\frac{3}{8}$

P(A ∩ B) =P(A). P(B) = $\frac{1}{8}$

P(A'∩B') = P($\bar{A}$). P($\bar{B}$) = $\frac{3}{8}$

(1–P(A)) (1–P(B)) = $\frac{3}{8}$

1–P(B) –P (A) + P(A) P (B) = $\frac{3}{8}$

1–(P(A) + P(B)) = $\frac{2}{8}$

P(A) + P(B)= $\frac{6}{8}$

$\frac{1}{P(B)} + P(B) = \frac{6}{8}$

P(B) = $\frac{1}{4}$

7. **(a)** P(A) = $\frac{3}{5}$, $P(B) = \frac{2}{5}$, P(C) = $\frac{3}{4}$

P(atleast hit by 2 shot) = P(A).P(B).P($\bar{C}$) +

P(A). P($\bar{B}$).P (C) +P($\bar{A}$).P (B).P (C) + P (A).P(B).P (C)

$= \frac{3}{5}\times\frac{2}{5}\times\frac{1}{4} + \frac{3}{5}\times\frac{3}{5}\times\frac{3}{4} + \frac{2}{5}\times\frac{2}{5}\times\frac{3}{4} + \frac{3}{5}\times\frac{2}{5}\times\frac{3}{4}$

$= \frac{6+27+12+18}{100}$

$= \frac{63}{100}$ = 0.63

8. **(b)**

x =	1	2	3	4	5	6	7	8
P(x)	0.15	0.23	0.12	0.10	0.20	0.08	0.07	0.05

E { x is a prime no.)

F = {x < 4}, P (E U F) = ?

E = {2,3,5,7}

F = {x: 1,2,3}

E U F = {1,2,3,5,7}

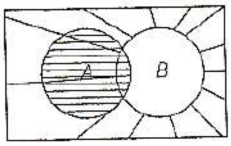
P (E U F) = 1 – {P(x = 4)+ P (x = 6)+ P (x = 8)}

= 1 – (0.10 + 0.08 + 0.05)

1 – (0.23)

0.77

9. **(b)**



$P(A') = 0.3, P(B) = 0.5$ and $P(A \cap B) = 0.3$

$P(A) = 1 - P(\bar{A}) = 1 - 0.3 = 0.7$

$P\left(\frac{B}{A\cup B'}\right) = \frac{P(B\cap(A\cup\bar{B}))}{P(A\cup\bar{B})} = \frac{P(A\cap B)}{P(A\cap\bar{B})}$

$P(A \cup \bar{B}) = 1 - P(B) + P(A \cap B)$

0.3 $= \frac{0.3}{1-0.5+0.3} = \frac{0.3}{0.8} = \frac{3}{8}$

10. **(b)**

When two dice are rolled, total no. of cases = 6 × 6 = 36

Let A is an event when sum of 5 occurs = $\{(1,4),(4,1),(2,3),(3,2)\}$

There are total 4 cases $P(A) = \frac{4}{36} = \frac{1}{9}$

Let B is an event sum 7 occurs

Total 6 cases

$P(B) = \frac{6}{36} = \frac{1}{6}$

Let C be the event when neither 5 nor 7 as a sum occurs

$= 1 - P(A) - P(B) = 1 - (1/9) - (1/6) = 26/36 = 13/18$

Now, the probability that 5 comes before 7 as a sum.

$P(A) + P(C) \times P(A) + P(C) \times P(C) + P(A) + ...$

$= \frac{P(A)}{[1-P(C)]} = \frac{1/9}{1-13/18} = \frac{2}{5}$

11. **(c)** Letters of word STATISTICS are A IICSSSTTT

Letters of word ASSISTANT are AAINSSSTT

Common letters in both are A, I, S and T

Probability of selecting A from both words $= \frac{1}{10}\times\frac{2}{9} = \frac{2}{90}$

Probability of selecting I from both words $= \frac{2}{10}\times\frac{1}{9} = \frac{2}{90}$

Probability of selecting S from both words $= \frac{3}{10}\times\frac{3}{9} = \frac{9}{90}$

Probability of selecting T from both words $= \frac{3}{10}\times\frac{2}{9} = \frac{6}{90}$

Probability of required event $= \frac{2}{90} + \frac{2}{90} + \frac{9}{90} + \frac{6}{90} = \frac{19}{90}$

12. **(d)** Bag contains 6 red and 4 green balls.

If outcome on dice is 1 then only 1 ball can be draw and probability that it will be red $= \frac{1}{6}\times\frac{6}{10}$

If outcome on dice is 2 then only 2 balls can be draw and probability that it will be red $= \frac{1}{6} \times \frac{{}^6C_2}{{}^{10}C_2}$

If outcome on dice is 3 then only 3 balls can be draw

and probability that it will be red $= \frac{1}{6} \times \frac{{}^6C_3}{{}^{10}C_3}$

If outcome on dice is 4 then only 4 balls can be draw

and probability that it will be red $= \frac{1}{6} \times \frac{{}^6C_4}{{}^{10}C_4}$

If outcome on dice is 5 then only 5 balls can be draw

and probability that it will be red $= \frac{1}{6} \times \frac{{}^6C_5}{{}^{10}C_5}$

If outcome on dice is 6 then only 6 balls can be draw

and probability that it will be red $= \frac{1}{6} \times \frac{{}^6C_6}{{}^{10}C_6}$

Required probability

$$= \frac{1}{6} \times \left[\frac{{}^6C_1}{{}^{10}C_1} + \frac{{}^6C_2}{{}^{10}C_2} + \frac{{}^6C_3}{{}^{10}C_3} + \frac{{}^6C_4}{{}^{10}C_4} + \frac{{}^6C_5}{{}^{10}C_5} + \frac{{}^6C_6}{{}^{10}C_6}\right] = \frac{1}{5}$$

13. (b) Let probability of getting odd number $= p$

Then probability of getting even number $= \frac{p}{3}$

In a dice, all outcomes getting $\{1,2,3,4,5,6\}$ are mutually exclusive and exhaustive cases hence

$P(1) + P(2) + P(3) + P(4) + P(5) + P(6) = 1$

$p + p/3 + p + p/3 + p + p/3 = 1$

Hence, $p = 1/4$

So, $P(\text{ odd number }) = p = 1/4$,

$P(\text{ even number }) = p/3 = 1/12$

$P(\text{ sum of number is even }) = P(\text{ both number are even })$

$+P(\text{ both number are odd})$

$= \frac{3}{12} \times \frac{3}{12} + \frac{3}{4} \times \frac{3}{4}$

$= \frac{5}{8}$

14. (a) Here we use Baye's theorem

Let E_1 denote the event that the letter came from TATANAGAR

And E_2 denote the event that the letter came from CALCUTTA.

Let A be the event that the two consecutive alphabets visible on the envelope are TA.

Since, the letters have come from either CALCUTTA or TATANAGER.

$P(E_1) = \frac{1}{2}, P(E_2) = \frac{1}{2}$

If E_1 has occurred then the letter has come from TATANAGAR. In the word TATANAGAR there are 8 consecutive alphabets

i.e. $\{\text{CA, AL, LC, CU, UT, TT, TA}\}$ and TA occurs once

$P\left(\frac{A}{E_2}\right) = \frac{1}{7}$

The probability that the letter has come from CALCUTTA is $P(E_2/A)$

By Baye's theorem,

$$\therefore P\left(\frac{E_2}{A}\right) = \frac{P(E_2)P\left(\frac{A}{E_2}\right)}{P(E_2)P\left(\frac{A}{E_2}\right) + P(E_1)P\left(\frac{A}{E_1}\right)}$$

$$= \frac{\frac{1}{2}\times\frac{1}{7}}{\frac{1}{2}\times\frac{1}{7}+\frac{1}{2}\times\frac{1}{4}} = \frac{4}{11}$$

15. (c) This is a total probability question.

We have 2 coins which have both heads (HH)

Hence, $P(\text{ lower face is head }) = 1$

We have one coin which have both tails (TT)

Hence $P(\text{ lower face is head }) = 0$

We have another 2 coins which are normal (HT)

Hence, $P(\text{ lower face is head }) = 1/2$

Probability that the lower face of the coin is a head.

$= \frac{2}{5} \times 1 + \frac{1}{5} \times 0 + \frac{2}{5} \times \frac{1}{2} = \frac{3}{5}$

16. (a) Let A be the event that both of them agree. T be the event that they both speak the truth and L be the event that they both lie.

Probability that both A and B speak the truth $= P(T) = xy$

Probability that both A and B lie

$= P(L) = (1-x)(1-y) = 1 - x - y + xy$

$$P\left(\frac{T}{A}\right) = \frac{P(T)P\left(\frac{A}{T}\right)}{P(T)P\left(\frac{A}{T}\right) + P(L)P\left(\frac{A}{L}\right)}$$

$$= \frac{xy \times 1}{(xy)1 + (1-x-y+xy)1} = \frac{xy}{1-x-y+2xy}$$

$$= \frac{xy}{1-x-y+xy+xy} = \frac{xy}{xy+(1-x)(1-y)}$$

17. (a)

$P(\overline{A \cup B}) = \frac{1}{6}, P(A \cap B) = \frac{1}{4}$ and $P(\bar{A}) = \frac{1}{4}$

$P(\overline{A \cup B}) = \frac{1}{6} = 1 - P(A \cup B)$

Hence, $P(A \cup B) = \frac{5}{6}$

$P(A) + P(B) - P(A \cap B) = \frac{5}{6}$

$\frac{3}{4} + P(B) - \frac{1}{4} = \frac{5}{6} \Rightarrow P(B) = \frac{1}{3}$

If A and B are independent events

$P(A \cap B) = P(A) \times P(B) = \frac{3}{4} \times \frac{1}{3} = \frac{1}{4}$

which is true, hence A and B are independent event.

18. (b) $P(1)=\frac{1}{6}, P(\overline{1})=\frac{5}{6}$

$P(A \text{ wins })=\frac{1}{6}+\frac{5}{6}\times\frac{5}{6}\times\frac{1}{6}+\cdots .=\frac{1}{6}\left[1+\frac{25}{36}+\left(\frac{25}{36}\right)^2+\cdots\right]$

$=\frac{\frac{1}{6}}{1-\frac{25}{36}}=\frac{\frac{1}{6}}{\frac{11}{36}}=\frac{6}{11}$

$P(B \text{ wins })=1-P(A \text{ wins })=1-\frac{6}{11}=\frac{5}{11}$

$E(\text{ amount of } A)=110\times\frac{6}{11}=60$

$E(\text{ amount of } B)=110\times\frac{5}{11}=50$

19. (c) Required probability $=P$ (atleast one person dies before 90yr) $\times P($ first person to die is $A_1)$

$=\left[1-\left(\frac{2}{3}\right)^4\right]\times\left[\frac{3!}{4!}\right]=\left(1-\frac{16}{81}\right)\times\frac{1}{4}=\frac{65}{324}$

20. (c) Probability that gun hit at first shot $=0.4$
Probability that gun hit at second shot $=0.3$
Probability that gun hit at third shot $=0.2$
Probability that gun hit at fourth shot $=0.1$
Probability that gun hits the plane
$=1-$ probability that none hit.
$=1-(1-0.4)(1-0.3)(1-0.2)(1-0.1)=0.6976$

21. (b) Since, there are just four matches to be played India can get a maximum of 8 points.
$P($ Indi get at least 7 points $)=P($ getting exactly 7 points $)+P($ getting exactly 8 points$)$
$=P$ (getting 2 in each of the 3 matches and 1 in one match $)+P($ getting 2 in each of the four matches$)$
$={}^4C_3(0.5)^3(0.05)+{}^4C_4(0.5)^4=0.025+0.0625$
$=0.0875$

22. (d) There will be two outcomes for such condition
{HTH, THT }
Sample space $=2\times2\times2=8$
Probability of required outcomes $=2/8=1/4$

23. (d) Let A be the number of coins showing heads
$P(A=50)=P(A=51)$
Using binomial distribution
$${}^{100}C_{50}p^{50}(1-p)^{50}={}^{100}C_{51}p^{51}(1-p)^{49}$$
$$\frac{p}{1-p}=\frac{51}{50}\Rightarrow p=51/101$$

24. (c) Using Possion distribution
Possion distribution is defined as
$f(x)=\frac{e^{-\lambda}\lambda^x}{x!}$

$P[x=3]=\frac{1}{4}P[x=4]\Rightarrow\frac{e^{-\lambda}\lambda^3}{3!}=\frac{1}{4}\frac{e^{-\lambda}\lambda^4}{4!}\Rightarrow\lambda=16$

$P[x=5]=kP[x=7]\Rightarrow\frac{e^{-\lambda}\lambda^5}{5!}=k\frac{e^{-\lambda}\lambda^7}{7!}$

$k\lambda^2=7\times6\Rightarrow k=\frac{42}{16\times16}=\frac{21}{128}$

25. (a) Sum of probabilities $=1$
$a+3a+5a+7a+9a+11a+13a+15a+17a=1$
$\Rightarrow 81a=1$
$\Rightarrow a=\frac{1}{81}$

26. (c) Refer to solution 12.

27. (b) The blades are in packet of 10 , so the number of blades in 10,000 packets are $=10000\times10=100000$
The probability for a razor blade is 0.002
So, the number of defective blades are $=100000\times 0.002=200$
So, the number of non-defective blades are $100000-200=99800$
So, the number of packets containing no defective blades are $=99800\div10=9980$

28. (b) A determinant of order 2 is of the form $\Delta=\begin{vmatrix} a & b \\ c & d\end{vmatrix}$
Its value is $ad-bc$
The total number of ways of choosing a,b,c and d is $2\times2\times2\times2=16$
If $\Delta\neq0$
If either $ad=1$ and $bc=0$ or $ad=0$ and $bc=1$
But $ad=1$ and $bc=0$ if and only if $a=d=1$ and at least one of b,c is zero.
Thus, $ad=1$ and $bc=0$ in three cases $[(a,b,c,d)$ $=(1,0,0,1)(1,0,1,1)(1,1,0,1)]$.
Similarly $ad=0$ and $bc=1$ in three cases $[(a,b,c,d)=(0,1,1,0)(1,1,1,0),(0,1,1,1)]$.
Thus, the probability of the required event is $\frac{6}{16}=\frac{3}{8}$

29. (a) For imaginary roots $b^2-4ac<0$
Total cases $=2\times2\times2=8$
For $b=1$ and a and c which can be chosen in 4 ways
For $b=2$ either $a=1,c=2$ or $a=2,c=1$ or $a=2,c=2$
Sample space $=\{(1,2,2)(1,1,2)(1,2,1)(1,1,1)(2,1,1)(2,2,1)(2,1,2)(2,2,2)\}$
Total number of favourable cases $=7$
Required probability $=\frac{7}{8}$.

30. (d) $P(A)=1,P(B)=\frac{1}{2}$ as $P(A)=1$ it is certain event
hence $P(A\cap B)=P(B)$

$P\left(\frac{A}{B}\right)=\frac{P(A\cap B)}{P(B)}=\frac{P(B)}{P(B)}=1$

$P\left(\frac{B}{A}\right)=\frac{P(A\cap B)}{P(A)}=\frac{P(B)}{P(A)}=\frac{1/2}{1}=\frac{1}{2}$

31. (a)

Step 1

Probability of getting k heads in n tosses

The probability of getting a head in a single toss is p = $\frac{1}{2}$

The probability of getting a tail in a single toss is

$1 - p = \frac{1}{2}$

The probability of getting exactly k heads in n tosses is

P(X = k) = $\binom{n}{k}\left(\frac{1}{2}\right)^k\left(\frac{1}{2}\right)^{n-k} = \binom{n}{k}\left(\frac{1}{2}\right)^n$

Step 2

We want to find the probability of getting 1, 3, 5.... heads

P(odd number of heads) $= \sum_{k \text{ odd } 0 \le k \le n}\binom{n}{k}\left(\frac{1}{2}\right)^n$

P (odd number of heads) = $\left(\frac{1}{2}\right)^n \sum_{k \text{ odd } 0 \le k \le n}\binom{n}{k}$

Step 3

We know that $(1+x)^n = \sum_{k=0}^{n}\binom{n}{k}x^k$

Set x = 1:

$(1+1)^n = \sum_{k=0}^{n}\binom{n}{k} = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \cdots.$

Set x = –1:

$(1-1)^n = \sum_{k=0}^{n}\binom{n}{k}(-1)^k = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} + \cdots..$

Subtract the second equation from the first

$2^n - 0 = 2\left(\binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \cdots ...\right)$

$\sum_{k \text{ odd } 0 \le k \le n}\binom{n}{k} = \frac{2^n}{2} = 2^{n-1}$

Step 4

P(odd number of heads) = $\left(\frac{1}{2}\right)^n . 2^{n-1} = \frac{2^{n-1}}{2^n} = \frac{1}{2}$

OR 2nd Method:

A fair coin tossed 'n' times

Total no. of outcomes = 2^n

Possible no. of outcomes of getting head in odd numbers of times

= 2^{n-1}

Probability that the head comes odd number of times

$= \frac{2^{n-1}}{2^n} = \frac{1}{2}.$

32. (d) Refer to solution 15.

33. (d) Probability that problem solved by $A = 1/2$

Probability that problem solved by $B = 1/3$

Probability that problem solved by $C = 1/4$

Probability that problem solved $= 1 - P$

(None of them solve the problem)

$= 1 - \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right) = \frac{3}{4}$

34. (b) Refer to solution 5.

35. (c) Probability that A hit the target is $4/5$.

Probability that B hit the target is $3/4$.

Probability that C hit the target is $2/3$.

Probability that target is hit at least 2 times.

$[AB\bar{C} + A\bar{B}C + \bar{A}BC + ABC$ (atleast two times)]

$\frac{4}{5} \times \frac{3}{4} \times \frac{1}{3} + \frac{4}{5} \times \frac{1}{4} \times \frac{2}{3} + \frac{1}{5} \times \frac{3}{4} \times \frac{2}{3} + \frac{4}{5} \times \frac{3}{4} \times \frac{2}{3} = 5/6$

36. (*)

As they are likely contradict to each other in stating the same fact only when one will speak truth and other will speak lie $= \frac{7}{10}\left(1 - \frac{6}{10}\right) + \left(1 - \frac{7}{10}\right) \times \frac{6}{10} = \frac{23}{50}$

37. (b) $P(A \cup B) = \frac{5}{6}, P(A \cap B) = \frac{1}{3}, P(B) = \frac{1}{2}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Hence, $P(A) = \frac{4}{6} = \frac{2}{3}$

For independent events

$P(A \cap B) = P(A) \times P(B)$

$1/3 = 2/3 \times 1/2$

Hence, A and B are independent events.

38. (b)

Clearly, $n(S) = (6 \times 6) = 36$

Let E = Event that the sum is a prime number

Then $E = \{(1,1), (1,2), (1,4), (1,6), (2,1), (2,3), (2,5), (3,2), (3,4), (4,1), (4,3), (5,2), (5,6), (6,1), (6,5)\}$ $n(E) = 15$

Hence, the probability that the total score is a prime number is

$\frac{15}{36} = \frac{5}{12}$

39. (b)

The probability of success is twice the probability of failure.

Let the probability of failure be p.

Probability of success $= 2p$

$p + 2p = 1$

⇒ 3p = 1

P $= \frac{1}{3}$

Hence probability of getting success is $2/3$ and failure is $1/3$.

Probability = Probability of getting atleast 4 success.

$= {}^6C_4\left(\frac{2}{3}\right)^4\left(\frac{1}{3}\right)^2 + {}^6C_5\left(\frac{2}{3}\right)^5\left(\frac{1}{3}\right) + {}^6C_6\left(\frac{2}{3}\right)^6$

$= \frac{496}{729}$

40. (b) Let probability of sale of brand 1 is $P(A)$.
Similarly, for brand 2 is $P(B)$
and brand 3 is $P(C)$.
Given, $P(A) = \frac{50}{100} = \frac{1}{2}$
$P(B) = \frac{30}{100} = \frac{3}{10}$
$P(C) = \frac{20}{100} = \frac{1}{5}$
Let W be the event that product required warranty means product is defective.
And also let the probability of required warranty work of brand 1 is $P(W/A)$.
Similarly for brand 2 is $P(W/B)$
and for brand 3 is $P(W/C)$
$\therefore P(W/A) = \frac{25}{100} = \frac{1}{4}$
$P(W/B) = \frac{20}{100} = \frac{1}{5}$
$P(W/C) = \frac{10}{100} = \frac{1}{10}$
Using total probability
Now, required probability
$= \frac{1}{2} \times \frac{1}{4} + \frac{3}{10} \times \frac{1}{5} + \frac{1}{5} \times \frac{1}{10}$
$= \frac{1}{8} + \frac{3}{50} + \frac{1}{50}$
$= \frac{50+24+8}{400} = \frac{82}{400} = 0.205$

41. (d) Suppose r integers, $0 < r < 10$, are chosen from $(0,1,\ 2, \dots,9)$ at random and with replacement.
Total number of possible outcomes $= 10^r$
No two integers are equal $= {}^{10}C_6$
The probability that no two are equal, is
$\frac{{}^{10}C_r}{10^r} = \frac{10}{10^r \times r! \times (10-r)!}$

42. (a) This is question of total probability.
As box contains 3 coins, 1 coin is fair hence probability of getting head or tail for this coin $P(H) = P(T) = \frac{1}{2}$
Second coin has two heads for this probability of getting heads $= 1$ and $P(T) = 0$.
Third coin is weighted probability of getting head is $1/3$ and getting tail is $2/3$.
Probability of choosing one coin $= 1/3$.
A coin is selected at random tossed then the probability that head appears $= 1/3(1/2 + 1 + 1/3) = 11/18$

43. (c) Probability of getting right answer $= \frac{1}{4}$
Probability of getting wrong answer $= \frac{3}{4}$
If a student attempt 5MCQ then the probability of getting atleast one correct answer $= 1 -$ Probability of no answer is correc
$= 1 - \left(\frac{3}{4}\right)^5 = 1 - \frac{243}{1024} = 0.7623$

44. (a) Probability of A being hit $= P(E) = 1 -$ (Probability of B missing) $\times$ (Probability of C missing)
$= 1 - \left(1 - \frac{1}{2}\right) \times \left(1 - \frac{1}{3}\right) = \frac{2}{3}$
Probability that A is hit by B and not $C = P[(B \cap \bar{C})/E]$
$= \frac{P(B) \times P(\bar{C})}{P(E)} = \frac{\frac{1}{2} \times \frac{2}{3}}{\frac{2}{3}} = \frac{1}{2}$

45. (c) Conditional $\leq P(A \cap B) \leq \min\{P(A), P(B)\}$
Min $P(A \cap B) =$ when both are independent $P(A \cap B) = P(A)P(B) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$
Max $P(A \cap B) =$ when smaller set is a subset of larger set. $P(A \cap B) = P(B) = \frac{1}{3} \Rightarrow$ Hence, $\frac{1}{6} \leq P(A \cap B) \leq \frac{1}{3}$

46. (d) The probability of getting number 1 in single throw of a dice $= \frac{1}{6}$
The probability of getting any number other than 1 in single throw of a dice $= \frac{5}{6}$
Now, we need to find the probability of getting number 1 when a fair dice is thrown and it appears in even numbered throws.
Probability that number 1 appears in second throw
$= \frac{5}{6} \times \frac{1}{6}$
Probability that number 1 appears in fourth throw
$= \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6}$
Probability that number 1 appears in sixth throw
$= \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6}$
And so it goes on infinity.
Thus, the probability of getting number 1 when a fair dice is thrown and it appears in even numbered throw
$= \left(\frac{5}{6} \times \frac{1}{6}\right) + \left(\frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6}\right) + \cdots \dots$
The above expression is a Geometric progression of infinite terms where first term $(a) = \frac{5}{36}$ and common ratio $(r) = \frac{25}{36}$
So, required probability $= S_\infty = \frac{\frac{5}{36}}{1 - \frac{25}{36}}$
Simplifying the expression, we get, $\Rightarrow S_\infty = \frac{5}{36-25} \Rightarrow$

$S_\infty = \frac{5}{11}$

Hence, probability of getting number 1 when a fair dice is thrown and it appears in an even numbered throw is $\frac{5}{11}$.

47. (d) There are total $2n+1$ tickets in which n tickets are even numbered and $n+1$ tickets are odd numbered.
Now, we have to select three tickets which are in AP.
Let a, b, c are numbers written on the 3 tickets which are drawn.
If these are in AP then $2b = a + c$ means $b = (a+c)/2$
a, b, c are natural numbers Hence b is also a natural number. Means $a + c$ must be divisible by 2 .
Hence, probability that 3 tickets drawn are in

AP $= \frac{{}^{n}C_2 + {}^{n+1}C_2}{{}^{2n+1}C_3}$

$= \frac{3n}{4n^2-1}$

48. (a) Box contains
2 blue caps, 4 red caps
5 green caps and 1 yellow cap
Total number of caps $= 12$
Sample space $= {}^{12}C_4$
For favourable cases $= {}^{7}C_4$
(None of them is green = total caps - green caps $= 12 - 5 = 7$)

Probability $= \frac{{}^{7}C_4}{{}^{12}C_4} = \frac{7}{99}$

49. (d) Let $n(A), n(B)$ and $n(A \cap B)$ denote the number outcomes favourable to A, B and $(A \cap B)$ respectively. If A and B are independent.

$P(A \cap B) = P(A)P(B)$

$\Rightarrow \frac{n(A\cap B)}{10} = \frac{4}{10} \times \frac{n(B)}{10}$

$\Rightarrow 5 \times n(A \cap B) = 2 \times n(B)$

Hence, $n(B) = \frac{5}{2} n(A \cap B)$

For independent events $n(A \cap B) \geq 1$
For different values of $n(A \cap B) = 2,4$ as $n(B)$ must be a natural number so $n(B) = 5,10$
As, we know $n(B) \leq 10$.

50. (c) We are given that $P(A \cup B) = 0.6$
and $P(A \cap B) = 0.3$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$0.6 = (1 - P(\bar{A})) + (1 - P(\bar{B})) - 0.3$

$P(\bar{A}) + P(\bar{B}) = 2 - 0.3 - 0.6 = 1.1$

51. (c) The probability of speaking truth of A is $P(A) = \frac{4}{5}$
The probability A speaks lie $P(\bar{A}) = 1 - \frac{4}{5} = \frac{1}{5}$ The probability of speaking truth of B is $P(B) = \frac{3}{4}$ The probability B speaks lie $P(\bar{B}) = 1 - \frac{3}{4} = \frac{1}{4}$ The probability of they contradict each other

$= P(A) \times P(\bar{B}) + P(\bar{A}) \times P(B)$

$= \frac{4}{5} \times \frac{1}{4} + \frac{1}{5} \times \frac{3}{4} = \frac{1}{5} + \frac{3}{20} = \frac{7}{20}$

52. (d) No. of houses $= 3 =$ No. of favourable cases
No. of applicants $= 3$
Total number of events $= 3^3$
(because each candidate can apply in 3 ways)
Required probability $= \frac{1}{3^3} + \frac{1}{3^3} + \frac{1}{3^3} = \frac{3}{3^3} = \frac{1}{9}$

53. (c) In total there are 5 horses, out of these Mr. A selects two horse. This can be done in ${}^{5}C_2 = 10$ ways.
Out of the five horses, only one horse wins.
Number of ways to select winning horse $= 1$
Number of ways to select one other horse out of 4 horses

$= {}^{4}C_1 = 4$

The probability that Mr. A selected the winning horse is

$= \frac{4}{10} = \frac{2}{5}$

54. (a) Refer to solution 8.

55. (b) Using Baye's theorem
We define the following events
A_1: He knows the answer
A_2: He does not know the answer
E: He gets the correct answer.
Thus, $P(A_1) = \frac{9}{10}$,

$P(A_2) = 1 - \frac{9}{10} = \frac{1}{10}, P\left(\frac{E}{A_1}\right) = 1$ and $P\left(\frac{E}{A_2}\right) = \frac{1}{4}$

Required probability

$$P\left(\frac{A_2}{E}\right) = \frac{P(A_2)P\left(\frac{E}{A_2}\right)}{P(A_1)P\left(\frac{E}{A_1}\right) + P(A_2)P\left(\frac{E}{A_2}\right)}$$

$$= \frac{\frac{1}{10}.\frac{1}{4}}{\frac{9}{10}.1 + \frac{1}{10}\frac{1}{4}} = \frac{1}{36+1} = \frac{1}{37}$$

56. (c) Let A be the event that the man reports that number 1 is obtained.
Let E_1 be the event that 1 is obtained and E_2 be its

complementary event.

Then, $P(E_1)$ = Probability that 1 occurs $= 1/6$

$P(E_2)$ = Probability that 1 does not occur

$= 1 - P(E_1) = 1 - 1/6 = 5/6$

Also, $P\left(\frac{A}{E_1}\right)$ = Probability that man reports 1 and it is actually a 1 $= 2/3$

$P\left(\frac{A}{E_2}\right)$ = Probability that man reports one and it is not a 1 $= 1/3$

By using Baye's theorem, probability that number obtained is actually a 1 .

$$P\left(\frac{E_1}{A}\right) = \frac{P(E_1)P\left(\frac{A}{E_1}\right)}{P(E_1)P\left(\frac{A}{E_1}\right)+P(E_2)P\left(\frac{A}{E_2}\right)}$$

$$= \frac{\frac{1}{6}\times\frac{2}{3}}{\frac{1}{6}\times\frac{2}{3}+\frac{5}{6}\times\frac{1}{3}} = \frac{2}{7}$$

57. **(a)** Refer to solution 9.

58. **(c)**

$P(E_2) = 0.35, P(E_1 \cup E_2) = 0.85$

and $P(E_1 \cap E_2) = 0.15$

$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

$\Rightarrow\ 0.85 = P(E_1) + 0.35 - 0.15 \Rightarrow P(E_1) = 0.65$

59. **(a)**

Mean $= np = 4$

{where (n) is the number of trails }

Variance $= npq = 2$

Let p is probability of wining and q is probability of losing.

$p + q = 1$

Dividing Eq. (ii) by Eq. (i), we get

$\frac{npq}{np} = \frac{2}{4} \Rightarrow q = \frac{1}{2}$

$\Rightarrow p + \frac{1}{2} = 1 \Rightarrow p = \frac{1}{2} \Rightarrow np = 4$

$\Rightarrow n \times \frac{1}{2} = 4 \Rightarrow n = 8$

Given that $P(x = 1)$

$P(x = 1) = {}^8C_1\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^7$

{Binomial distribution $= {}^nC_r(p)^r(q)^{n-r}$}

$8 \times \frac{1}{2} \times \frac{1}{2^7} = 2^3 \times \frac{1}{2^8} = \frac{1}{2^5} = \frac{1}{32}$

60. **(b)**

$P(A \cup B) = P(A) + P(B) - (A \cap B)$

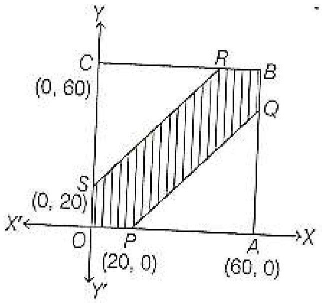
$\frac{5}{6} = P(A) + \frac{1}{2} - \frac{1}{3} \Rightarrow P(A) = \frac{2}{3}$

If A and B are independent to each other

$P(A \cap B) = P(A) \times P(B) = \frac{2}{3} \times \frac{1}{2} = \frac{1}{3}$

Hence, A and B are independent events.

61. **(a)** Let A and B arrive at the place of their meeting x minutes and y minutes after 6pm.



Their meeting is possible only if $|x - y| \leq 20$

$OABC$ is a square, where

$A = (60,0), C = (0,60)$

Considering the equality $|x - y| = 20$

The area representing the favourable cases

= Area $OPQBRSO$

= Area of square $OABC$ − Area of $\triangle PAQ$

Area of $\triangle SRC$

$= (60)(60) - (1/2)(40)(40) - (1/2)(40)(40)$

$= 3600 - 1600 = 2000$ sq unit

Total way = Area of square $OABC$

$= (60)(60)$

= 3,600 sq. unit

Required probability $= \frac{2000}{3600} = \frac{5}{9}$.

62. **(a)** Total number of ways of choosing 3 numbers from the given set of numbers 1,2 , 99 = ${}^{99}C_3$.

Let us divide the given numbers into 3 group G_1, G_2, G_3 as follows

G_1: 3,6,9, … . .99

[33 elements]

G_2: 1,4,7, … . .97

[33 elements]

G_3: 2,5,8, … 98

[33 elements]

Given that $a^3 + b^3 + c^3 - 3abc$ is to be divisible by 3 .

If $a^3 + b^3 + c^3 - 3abc$ is divisible by 3 , then which is possible in the following cases.

(ii) All the number belong to the second group.

(iii) All the number belong to the third group.

(iv) One number belong to the first group, one belong to second group and one belong to the third group.

So, favourable number of cases are $3 \cdot {}^{33}C_3 + ({}^{33}C_1)^3$

Required probability $= \frac{3 \cdot {}^{33}C_3 + ({}^{33}C_1)^3}{{}^{99}C_3}$

63. **(b)** There are A and B have to select as number from set of numbers $\{1,2, \ldots .25\}$.

According to given condition, they will win prize if they choose the same numbers, i.e, then probability of choosing same numbers is $1/25$.

Hence, the probability that they will not win a prize in a single trail

$= 1 - \frac{1}{25} = \frac{25-1}{25} = \frac{24}{25}$

64. **(a)** We have, first 30 natural numbers.

Two numbers are selected a and b such that $a^2 - b^2$ is divisible by 3 .

$n(S) = {}^{30}C_2$

$a^2 - b^2$ is divisible by 3 .

$(a+b)(a-b) = 3$ m

If $(a-b)$ is multiple of 3 , then $a^2 - b^2$ is multiple of 3 .

Such numbers are

$(1,4,7, \ldots \ldots .28)(2,5, \ldots ., 29), (3,6,9, \ldots .30)$ are possible.

${}^{10}C_2 + {}^{10}C_2 + {}^{10}C_2 = 45 \times 3 = 135$

$(a+b)$ is multiple of 3 then $a^2 - b^2$ is multiple of 3 .

Such numbers are possible

$\{(1,2)(1,5), \ldots .1,29\} \cdot \{(4,2),(4,5) \ldots \ldots .\} \ldots . .\{(28,2) \ldots . .28,29\}$

Total $10 \times 10 = 100$

Total numbers are $135 + 100 = 235$

Required probability $= \frac{235}{{}^{30}C_2} = \frac{235 \times 2}{30 \times 29} = \frac{47}{87}$

65. **(d)** As, $0.4 + 0.6 = 1$, the man either takes a step forward or a step backward.

Then, the probability of success in one step

$p = 0.4 = 2/5$

The probability that one step backward

$= p = 0.6 = 3/5$

In 11 steps he will be one step away from the starting point if the numbers of forward and back ward differ by 1 .

So, the number of forward $= 6$

The number of back ward $= 5$

or the number of forward $= 5$,

The number of backward $= 6$

The required probability

$= {}^{11}C_6 \left(\frac{2}{5}\right)^6 \cdot \left(\frac{3}{5}\right)^5 + {}^{11}C_5 \left(\frac{2}{5}\right)^5 \cdot \left(\frac{3}{5}\right)^6$

$= 462 \times (0.24)^5$

66. **(b,d)**

(a) Option If $P(U) = P(V)$, then $n(U) = n(V)$, this does not imply that $U = V$ hence this option is false.

(b) Option If $P(U) = Q, P(\bar{U}) = 1 - 0 = 1$, hence $U^c = S$ (this option is true)

(c) If $U \cap V = \phi$, then U and V are mutually exclusive, not independent. Hence False.

(d) If U and V are independent, then U^c and V^c are also independent. (True)

67. **(a)** $P($ win of $5000) = 0.001, P($ win $= 2000) = 0.003$

Hence, there will be a case that the man does not win any prize.

Fair prize is defined as the case expected gain $= 0$

Then expected gain $=$ expected winning - expected loss

$= 0$

F is the fair prize

$= 5000 \times P($ win $= 5000) + 2000 \times P($ win $= 2000) - F \times P($ win $= 0)$

$=0$

$= 5000 \times 0.001 + 2000 \times 0.003 - (1 - 0.001 - 0.003) \times F = 0$

$F \times (0.9966) = 11$

$\Rightarrow F = \frac{11}{0.9966} \cong 11$

68. **(c)** Let $x = P$ (computer tums out to be defective, given that it is produced in plant T_2).

$\Rightarrow x = P\left(\frac{D}{I_2}\right)$

where, $D =$ defective computer.

P (computer turns out to be defective given that is produced in plant $T_1) = 10x$

i.e.,

$P\left(\frac{D}{T_1}\right) = 10x$ .... (i)

Also, $P(T_1) = \frac{20}{100}$ and $P(T_2) = \frac{80}{100^\circ}$

Given, $P($ defective computer $) = \frac{7}{100}$

i.e., $P(D) = \frac{7}{100}$ .... (ii)

Using law of total probability

PD= $P(T_1) - P\left(\frac{D}{T_1}\right) + P(T_2) \cdot P\left(\frac{D}{T_2}\right)$

$\frac{7}{100} = \left(\frac{20}{100}\right) - 10x + \left(\frac{80}{100}\right) \cdot x$

$7 = (280)x \Rightarrow x = \frac{1}{40}$ .... (iii)

$\therefore P\left(\frac{D}{T_2}\right) = \frac{1}{40}$ and $P\left(\frac{D}{T_1}\right) = \frac{10}{40}$

$\Rightarrow P\left(\frac{\bar{D}}{T_2}\right) = 1 - \frac{1}{40} = \frac{39}{40}$ and $P\left(\frac{\bar{D}}{T_1}\right) = 1 - \frac{10}{40} = \frac{30}{40}$ ...

Using Baye's theorem,

$P\left(\frac{T_2}{\bar{D}}\right) = \frac{P(T_2) \cdot P\left(\frac{\bar{D}}{T_2}\right)}{P(T_1) \cdot P\left(\frac{\bar{D}}{T_1}\right) + P(T_2) \cdot P\left(\frac{\bar{D}}{T_2}\right)}$

$= \frac{\frac{80}{100} \cdot \frac{39}{40}}{\frac{20}{100} \cdot \frac{30}{40} + \frac{80}{100} \cdot \frac{39}{40}} = \frac{78}{93}$

69. **(c)** Probability that A will solve the problem $= 1/2$

Probability that A will not solve the problem $= 1/2$

Probability that B will not solve the problem $= 1/4$

Probability that B will solve the problem $= 3/4$

Let probability that C will solve the problem $= p$

Then probability that C will not solve the problem

$= 1 - p$

The probability that problem is solve $= 1 -$ Probability that no one solve the problem.

$63/64 = 1 - P(\bar{A})P(\bar{B})P(\bar{C}) = 1 - (1/2)(1/4)(1-p)$

p= 7/8

70. **(b)** Refer to solution 55.

71. **(d)** Probability of getting triplet when 3 dice are thrown

$= \frac{6}{6^3} = \frac{1}{36}$

Probability of not getting triplet when 3 dice are thrown

$= 1 - \left(\frac{1}{36}\right) = \frac{35}{36}$

Here, we use binomial distribution.

Probability of throwing triplet not more that twice = Probability (0 triplet) + Probability (1 triplet) + Probability (2 triplets) $= 1 -$ Probability of getting 3 triplets

$\Rightarrow 1 - {}^3C_3\left(\frac{1}{36}\right)^3 \Rightarrow 1 - \frac{1}{36^3}$

72. **(c)**

$P(E) = 0.25, P(F) = 0.50$

$P(E \cap F) = 0.14$

$P(\bar{E} \cap \bar{F}) = P(\overline{E \cup F})$

$= 1 - P(E \cup F) = 1 - P(E) - P(F) + P(E \cap F)$

$= 1 - 0.25 - 0.50 + 0.14 = 0.39$

73. **(d)** Probability that a man aged x dies in a year $= p$

Therefore the probability that at least one man dies in a year $= 1 - p$

No man dies in a year $= 1 - (1-p)^n$

Probability that out of n men A_1 is first to dies $= \frac{1}{n}$

Hence, the probability that A_1 dies in the one year

$= \frac{1}{n}[1 - (1-p)^n]$

74. **(a)** Simple space of selecting a number x from

1,2,3 ....100 $= {}^{100}C_1 = 100$

Solving the inequality $x + \frac{100}{x} > 29$

$\frac{x^2 - 29x + 100}{x} > 0, \frac{(x-25)(x-4)}{x} > 0$

$0 < x < 4$ Or $x > 25$ Favourable cases $=$ [1,2,3,26, ... 100] $= 78$ number.

Probability $= \frac{78}{100} = \frac{39}{50}$

75. **(d)** $x^2 + y^2 - 2y = 0$ centre $(0,1)$ radius $(r) = 1$

$x^2 + y^2 - 2y - 3 = 0$ centre $(0,1)$ radius $(R) = 2$

Sample space $\pi R^2 = \pi \times 2^2 = 4\pi$

Then, the probability that the point has been taken from smaller circle is

Favourable cases $= \pi r^2 = \pi \times 1^2 = \pi$

Probability $= \frac{\pi}{4\pi} = \frac{1}{4}$

76. **(c)** $P(A \cap B) = P(A)P(B)$, hence A and B are independent events.

Option (a) $P\left(\frac{B}{A}\right) = P(B)$ and option

(d) $P\left(\frac{A}{B}\right) = P(A)$

Hence, (a) and (d) options are wrong.

Option (c) $P(A \cup B)^c = P(A^c)P(B^c)$

Hence, option (c) is correct.

77. **(c)** A four digit number is formed using the digits 1,2, 3,

4,5 without repetition

Sample space $= {}^5P_4 = 5! = 120$

The probability that number is divisible by 3 .

Any number divisible by 3 if sum of its digit is divisible by 3 .

Hence, if we choose 4 digits 1,2,4,5.

Sum of its digits is 12 . Favourable cases = 4 !

Probability $= \frac{4!}{5!} = \frac{1}{5}$

78. **(c)** Refer to solution 60.

79. (a) Probability A speaks truth $P(AT) = \frac{60}{100} = \frac{3}{5}$

$\Rightarrow$ Probability A speaks lie $P(AL) = \frac{40}{100} = \frac{2}{5}$

Probability B speaks truth $P(BT) = \frac{50}{100} = \frac{1}{2}$

$\Rightarrow$ Probability B speaks lie $P(BL) = \frac{50}{100} = \frac{1}{2}$

Probability that they contradict each other stating the same fact $= P(AT \cap BL) + P(AL \cap BT)$

$= \frac{3}{5} \times \frac{1}{2} + \frac{2}{5} \times \frac{1}{2} = \frac{1}{2}$

80. (a) $P(A) = \frac{2}{3}$

$P(A \cup B) = \frac{3}{4}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\frac{3}{4} = \frac{2}{3} + P(B) - P(A \cap B)$

$P(A \cap B) = P(B) - \frac{1}{12}$

$0 \le P(A \cap B) \le P(A)$

$0 \le P(B) - \frac{1}{12} \le \frac{2}{3}$

$\frac{1}{12} \le P(B) \le \frac{2}{3} + \frac{1}{12} \Rightarrow \frac{1}{12} \le P(B) \le \frac{3}{4}$

81. (a) $5Y + 4B + 3G = 12$ balls

Non black ball = 8 balls

$P($ No black ball is selected $) = \frac{^8C_3}{^{12}C_3} = \frac{8 \times 7 \times 6}{12 \times 11 \times 10} = \frac{14}{55}$

82. (b)

Bag A	Bag B
4G	5G
5B	3B
6W	2W

B : Black ball is drawn from bog A

G: Green ball drawn from bag B

$P\left(\frac{B}{G}\right) = \frac{P(B \cap G)}{P(G)}$

$= \frac{\frac{5}{15} \times \frac{5}{11}}{\frac{5}{15} \times \frac{5}{11} + \frac{4}{15} \times \frac{6}{11} + \frac{6}{15} \times \frac{5}{11}}$

$= \frac{25}{79}$

83. (c)

$x^2+y^2<4$ P probability of (r,s) r > s

P=(r,s) first quadrant satisfying

P(E)= 1/2

84. (c) Total thrown = 8

Odd no. of thrown = (1,3,5,7) = 4

P(E) of getting head in odd thrown = ½

85. (b)

Total patient = 3 (A,B, C)

P(E) of recovering a patient = 0.6

P($\bar{E}$) not recovering a patient = 0.4

P(E) = P(A). P (B) P($\bar{C}$) + P (A). P($\bar{B}$). P(C) +P($\bar{A}$).P(B). P(C)

$\Rightarrow$ (0.6 x 0.6 x 0.4) + (0.6 x 0.4 x 0.6) + 0.9 x 0.6 x 0.6

= 3 x 0.36 x 0.4

= 0.432

86. (c) Total no. divided by 2 and 3

Both = 6,12,18….96

Total = 16

Total natural no. = 100

P(E) = $\frac{^{16}C_3}{^{100}C_3}$

$\frac{16 \times 15 \times 14}{100 \times 99 \times 98}$

$\frac{4}{1155}$

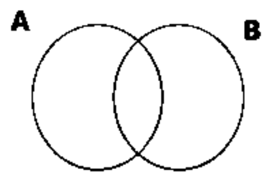
87. (d)

P ((A $\cap$B^C) $\cup$ (A^C$\cap$B))

$\Rightarrow$ P (A $\cup$ B) – P (A $\cap$B)

= P (A) + P (B) – P (A $\cap$ B) – P (A $\cap$ B)

= P (A) + P (B) – 2P (A $\cap$ B)



88. (b) P(A) = 40% = 4/10

P(B) =50% = 5/10

P(A).P($\bar{B}$) +P(B) P($\bar{A}$)

$\frac{4}{10} \times \frac{5}{10} + \frac{5}{10} \times \frac{6}{10}$

$\frac{50}{100} = \frac{1}{2}$

89. (b)

Total ways to arrange 5 captains in a line:

5! = 120

If I and A are together, consider them as one "block". Then, we have:

- The I–A block
- The other 3 captains

So total "items" to arrange: 1 + 3 = 4

Number of ways to arrange these 4 items:

4! = 24

Within the I–A block, I and A can be in 2 orders: (I, A) or (A, I).

So total favorable arrangements:

24 × 2 = 48

Probability

$P($I and A together$) = \frac{\text{favorable}}{\text{total}} = \frac{48}{120} = \frac{2}{5}$

Ans. $\frac{2}{5}$

90. (d)

Given:

P(E) = P(E/F)

This means:

P(E I F) = P(E) $\Rightarrow$ $P(E \cap F) = P(E)\,P(F)$

Now check each option:

(a) P(F) = P(F I E) ⇒ also independence

(b) E and F are independent

(c) E^C and F are independent ⇒ true if E and F are independent

(d) $P(E^C)P(F^C) \neq P(E^C \cap F^C)$ ⇒ complements not independent ⇒ E, F not independent

Ans. (d)

91. (a)

1. Total outcomes = 4 × 4 = 16
2. A : x ≥ 2 → 12 outcomes → 3/4
3. B : y > x → 3 + 2 + 1 = 6 outcomes → 3/8
4. A ∩ B : from above, only x = 2, 3 work → 2 + 1 = 3 outcomes → 3/16 ≠ 1/4

Answer: (a) is NOT true.

92. (a)

Probability Calculation

The probability that a randomly selected student scored between 450 and 500 is calculated using the properties of the normal distribution and Z-scores.

Step 1: Convert Scores to Z-scores

The Z-score is calculated using the formula Z = $\frac{X-\mu}{\sigma}$, where X is the score, μ is the mean, and σ is the standard deviation.

For X = 500 : $Z_{500} = \frac{500-500}{100} = 0$

For X = 450 : $Z_{450} = \frac{450-500}{100} = \frac{-50}{100} = -0.5$

Step 2: Determine the Probability Using Cumulative Distribution

The probability P $(450 \leq X \leq 500)$ is equivalent to $P(-0.5 \leq Z \leq 0)$.

This probability can be expressed as

$P(Z \leq 0) - P(Z \leq -0.5)$.

It is known that for a standard normal distribution, $P(Z \leq 0) = 0.5$

Using the symmetry property of the standard normal distribution,

$P(Z \leq -0.5) = 1 - P(Z \leq 0.5)$

Given that the cumulative distribution of the standard normal random variable at 0.5 is 0.691, it follows that P(Z ≤ 0.5) = 0.691

Therefore, P(Z ≤ 0.5) = 1 - 0.691 = 0.309

Step 3: Calculate the Final Probability

The probability P(-0.5 ≤ Z ≤ 0) is calculated as:

P(-0.5 ≤ Z ≤ 0) = P(Z ≤ 0) - P(Z ≤ -0.5) = 0.191.

Final Answer

The probability that a randomly selected student scored between 450 and 500 is 0.191.

93. (b)

Mean accidents per week = 2

* For **2 weeks**, Poisson mean:

λ = 2 × 2 = 4

Probability of exactly 3 accidents in 2 weeks:

$P(X = 3) = \frac{\lambda^3 e^{-\lambda}}{3!} = \frac{4^3 e^{-4}}{3!}$

Given $P(X = 3) = ke^{-6}$, equate:

$\frac{64}{6}e^{-4} = ke^{-6}$

Divide both sides by e^{-6}

$k = \frac{64}{6}e^2$

Since $e^2 \approx 7.389$

$k = \frac{64}{6} \times 7.389 \approx 36$

94. (c)

Probability Calculation

The probability of choosing the blue coin is $\frac{1}{2}$.

The probability of choosing the red coin is $\frac{1}{2}$.

The probability of getting a head with the blue coin is 0.99.

The probability of getting a head with the red coin is 0.01.

The probability of getting a head is calculated using the law of total probability:

P(Head) = P(Head/Blue) × P(Blue) + P(Head/Red) × P(Red)

Substituting the values, the probability of getting a head is calculated as

$0.99 \times \frac{1}{2} + 0.01 \times \frac{1}{2}$

This simplifies to $\frac{0.99+0.01}{2} = \frac{1.00}{2} = 0.5$

Final Answer

The probability of getting a head is 0.5.